# Malvern Madondo


CONTACT INFORMATION

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Google Scholar Profile


## EDUCATION

**Emory University**, Atlanta, GA *August, 2019 - present*
PhD Student, Computer Science and Informatics

**The College of St. Scholastica**, Duluth, MN *August, 2015 - May, 2019*
B.S., Computer Science/Information Systems and B.A., Mathematics

## TECHNICAL SKILLS

**Programming Languages:**

- Most experience: Java, Python(incl. TensorFlow, Keras, PyTorch), MATLAB
- Some Experience: PHP/Hack, JavaScript, React, SQL, HTML/CSS, Swift

**Version Control:** Git, Bitbucket, Mercurial
**Development:** Cross-platform(Native Templates, PHP/Hack, React); iOS (Swift); Android (Java),
**Operating Systems:** macOS, Linux, Windows
**Methodologies:** Agile (SCRUM) and Waterfall

## RESEARCH EXPERIENCE

### Reinforcement Learning for Mean Field Games

*Machine Learning and Inverse Problems (MLIP), Emory University* *June, 2020 - present*

- Formulating mean field games and control problems from a reinforcement learning (RL) perspective. Mentored by Dr. Lars Ruthotto.
- Studying RL-based approaches to solving adaptive optimal control problems, particularly in systems with nonlinear, unknown, and stochastic dynamics.

### Towards Mastery Learning in Computer Science Education

*Enhanced Learning and Instructional Technologies at Emory (ELITE)* *June, 2020 - present*

- Mentored by Dr. Davide Fossati, I proposed a method to utilize student-authored question-answer pairs to create an "unlimited" (or very large) supply of practice problems.
- Proposal incorporated in CS170 class. Next steps will be analyzing submitted question-answer pairs, annotate them by skill, and assess for mastery of concepts.

### UPenn and Mayo Clinic's Seizure Detection Challenge

*CS534 Machine Learning, Emory University* *February - May, 2020*

- Worked in a team of three to detect seizures in intracranial EEG recordings
- Developed per-subject ensemble-based models that predicted whether a given clip, collected from 8 human patients with epilepsy and 4 canines with naturally-occurring epilepsy, is a seizure or not (ictal or interictal).
- Computed and contrasted the mean area under the ROC curve of boosting methods with that obtained from prediction results of CNN and SVM models.

### OpenAI BipedalWalker-v2 Challenge

*CS557 Artificial Intelligence, Emory University* *October - December, 2019*

- Worked in a team of two to train a 2D bipedal walker to navigate rough terrain.
- Implemented and applied three Reinforcement Learning strategies: 1) Deep Deterministic Policy Gradients (DDPG); 2) Evolution Strategies (ES); and 3) Augmented Random Search (ARS).
- Solving the environment required an average reward of 300 over 100 consecutive trials. Solved the challenge using ARS in about 1000 episodes (and lots of tea).

## Internship Experience

**Facebook -** Crisis Response Team, Menlo Park, CA

**Software Engineering Intern** *May - August, 2018*

- Worked with the Crisis Response/Safety Check team. Full-stack and back-end work in Hack, XHP, and JavaScript.
- Worked on bringing Community Help, which allows users to find and offer help during a crisis, to Facebook's mobile site.
- Built a Mark Safe QP on WWW platform that resulted in a 25% increase in primary clicks, 43% increase in QP dismissals and 50% increase in 'Does Not Apply' clicks.

**Facebook -** FBU, Menlo Park, CA

**Software Engineering Intern** *June - August, 2017*

- Participated in the Facebook University for Engineering program – iOS Development track.
- Developed from scratch an iOS app that integrated location-based facts, Tinder-style fact sharing, localization, image recognition (utilizing IBM Watson API) and Wikipedia descriptions.

**The College of St. Scholastica -** Marketing Department, Duluth, MN

**Web Assistant** *October, 2016 - May, 2019*

- Supported the web and marketing goals of the Interactive Services Team through web content design, creation, and editing, while following branding guidelines and best practices.
- Prepared site pages and content to meet accessibility guidelines.

**State of Minnesota -** Department of Human Services , St. Paul, MN

**Urban Scholar and Management Analyst** *May - August, 2016*

- Assisted the Member Provider Services communications team in the development and implementation of a process to archive Provider Manual web page revisions and index web page documents.
- Implemented the Minnesota State Government's Plain Language initiative in web page revisions to promote accessibility.
- Participated in the Urban Scholars Leadership Institute. Awarded Certificate of Commendation.

**The College of St. Scholastica -** IT Help Desk, Duluth, MN

**Team Lead and Tech Assistant** *September, 2015 - October, 2019*

- Co-led and supervised a team of 15 Tech Assistants offering technical support, troubleshooting and diagnosing network and technical issues.
- Demonstrated advanced customer service skills while interacting and communicating with peers, faculty, and staff from diverse backgrounds.

## Publications

Madondo, M. and Gibbons, T. **Learning and Modeling Chaos Using LSTM Recurrent Neural Networks.** *Midwest Instruction and Computing Symposium (MICS 2018).*
`http://micsymposium.org/mics2018/proceedings/MICS_2018_paper_26.pdf`

Madondo, M., Gomez, DM., and Ciernia, N. **Hitchhiker's Guide to Computer Science for Social Good** *Midwest Instruction and Computing Symposium (MICS 2018).*
`http://micsymposium.org/mics2018/proceedings/MICS_2018_paper_41.pdf`.

## Poster Presentations

Madondo, M., Vang, J., and Gibbons, T. Judging a whale by its tail: A Kaggle Humpback Whale Identification Challenge. *MICS 2019*, NDSU, Fargo, North Dakota.

## Honors and Awards

2020 Science ATL Communication Fellow

2020 Computing Research Association (CRA-WP) Grad Cohort for Underrepresented Minorities and Persons with Disabilities (URMD) scholarship recipient

2019 The College of St. Scholastica: graduated summa cum laude

2019 Webster Scholar (Honors program), The College of St. Scholastica

2018 Minnesota High Tech Association Foundation (MHTF) scholarship recipient

2018 Affirm scholarship recipient for the ACM Richard Tapia Conference

2018 Quip Diversity Scholarship runner-up

2018 Student Employee of the Year Award, The College of St. Scholastica

2017 Code2040 Tech Trek fellow

2017 Minnesota Private Colleges, Phillips Scholarship recipient

2016 Minneapolis Urban Scholars Leadership Award

2015-2019 Philanthropic Ventures Foundation Grace Scholarship recipient

## Teaching Experience

**Emory University**, Atlanta, GA

**Instructor** *August, 2020 - present*

- Co-teaching CS 170(Introduction to Computer Science) online.
- Lead weekly lab sessions and involve students in more hands-on learning activities.

**Teaching Assistant - CS 171** *January - May 2020*

- Assisted students in Dr. Hashemi's CS171 course with questions during lectures and office hours.
- Course emphasized the use and implementation of data structures, fundamental algorithms with introductory algorithm analysis, and object oriented design and programming with Java.

**Teaching Assistant - CS 170** *August - December 2019*

- Supported students taking the Intro to CS course by providing 1-on-1 instruction during lectures and office hours.
- Graded assignments and weekly quizzes. Topics included: fundamental computing concepts, general programming principles, the Unix Operating System, and the Java programming language.

**The College of St. Scholastica**, Duluth, MN

**Math Teaching Assistant** *October, 2018 - May 2019*

Held weekly office hours to support students taking Calculus courses.

**Google IgniteCS Program -** Lincoln Park Middle School, Duluth, MN

**Mentor** *January, 2018 - May 2018*

Mentored Middle School students and taught weekly course in Computational Thinking as a foundation for Computer Science.

**CodePath.Org -** CSS, Duluth, MN

**Technical Program Manager (iOS Development Tutor)** *September, 2017 - May 2019*

Provided technical support for students during labs and assignments. Lead tutorials and manage in-class activities.

## Leadership and Other Activities

**Emory University** — Computer Science and Informatics Department, Atlanta, GA
**Student Ambassador** *September, 2019 - present*
Bridge the gap between CSI students and faculty. Also plan student events and other activities.

**NORTHFORCE** — Mentor Connection Program, Duluth, MN
**Student Ambassador** *October, 2016 - May, 2019*
Connect students with local professionals and share about networking opportunities.

**Marketing Department** — The College of St. Scholastica, Duluth, MN
**Blogger** *September, 2015 - May, 2019*
Maintained department's social media presence to engage current and prospective students. Provided insights into student life while sharing experiences as a student.

**WellU Program** — The College of St. Scholastica, Duluth, MN
**Storm's Advocate** *August, 2016 - May, 2018*
Present educational programs and table at events promoting a lifestyle of health and wellness to college community.

**DIGI-KEY Electronics** — Programming Competition, Thief River Falls, MN
**Competitor** *October, 2017*
Mentored by professor Kris Glesener and in a team of four, placed fifth place overall. Won first place in two "blender" sessions.

**Office of International Programs** — The College of St. Scholastica, Duluth, MN
**Student Ambassador** *August — September, 2016*
Assisted in the organization and implementation of the summer orientation program for incoming international students.

**Youth Theology Institute** — The College of St. Scholastica, Duluth, MN
**Peer Leader** *July, 2016*
Mentored participants and led opportunities for community building and small group discussions.

**Midwest Instruction & Computing Symposium** — Robotics Competition
**Competitor** *April, 2016 and April, 2017*
Designed and programmed an autonomous robot that could trace a path and play a 3-hole miniature golf course, with the help of two other students and guidance from Dr. Tom Gibbons.

**Mathematical Association of America, NCS** — Team Competition
**Competitor** *November, 2015-2018*
Participated in the annual math competition organized by the Mathematical Association of America, North Central Section. Problem level ranged from moderately easy to difficult Putnam level Math problems.

## Organizations

Society for Industrial and Applied Mathematics (Emory SIAM Chapter)
Machine Learning and Data Science (MLDS) - Africa Network
United States Student Achievers Program (USAP)

## Languages

English (Full Proficiency)
Shona (Bilingual/Native Proficiency)